



PLUGGING INTO THE FUTURE:
AN EXPLORATION OF ELECTRICITY
CONSUMPTION PATTERNS
Project Based Experiential Learning Program

Handbook

SmartBridge Educational Services Pvt. Ltd.

BY

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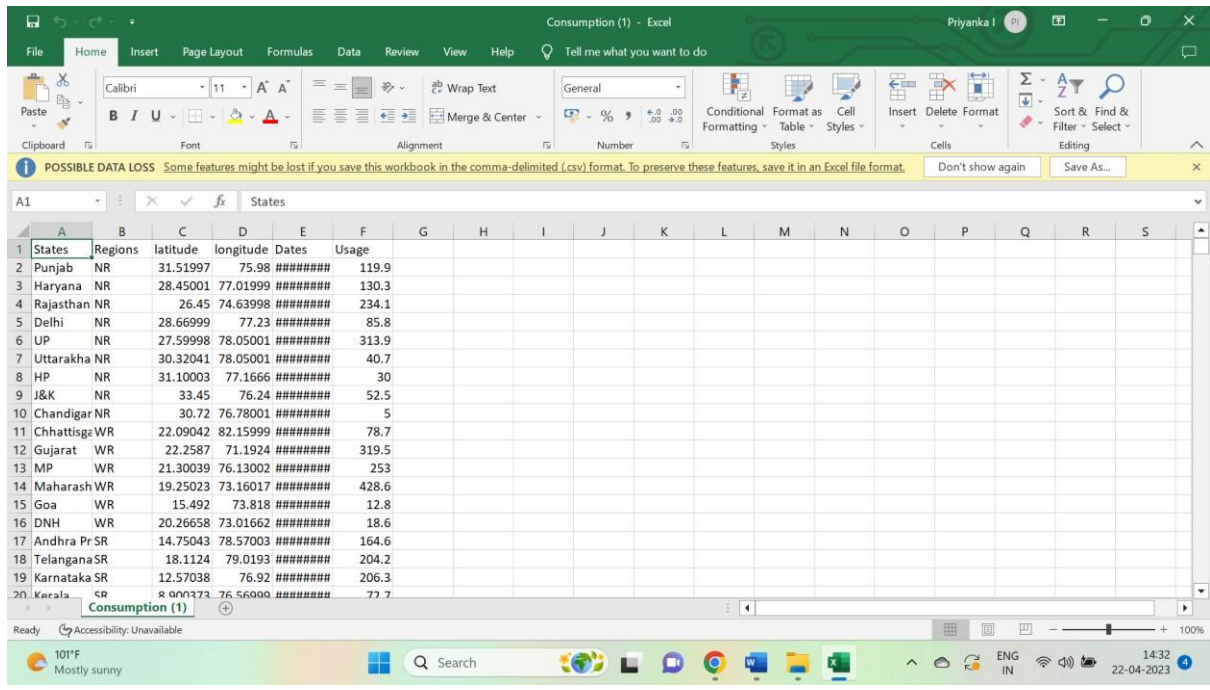
Sri Venkateswara College of Engineering

○INTRODUCTION

India is the world's third-largest producer and third largest consumer of electricity. The national electric grid in India has an installed capacity of 370.106 GW as of 31 March 2020. Renewable power plants, which also include large hydroelectric plants, constitute 35.86% of India's total installed capacity. During the fiscal year (FY) 2019–20, the total electricity generation in the country was 1,598 Terawatt-hour, of which 1,383.5 Terawatt-hour generated by utilities. The gross electricity consumption per capita in FY2019 was 1,208 kWh. In 2015-2016, electric energy consumption in agriculture was recorded as being the highest (17.89%) worldwide. The per capita electricity consumption is low compared to most other countries despite India having a low electricity tariff. In light of the recent COVID-19 situation, when everyone has been under lockdown for the months of March to June the impacts of the lockdown on economic activities have been faced by every sector in a positive or a negative way. The dataset is exhaustive in its demonstration of energy consumption state wise. Analysing Electricity Consumption in India from Jan 2019 till 5th December 2020. This dataset contains a record of electricity

consumption in each state of India, here we are going to analyse State wise, Region wise and Overall Electricity consumption in India.

ODATA EXTRACT AND COLLECTION



States	Regions	latitude	longitude	Dates	Usage
Punjab	NR	31.51997	75.98	#####	119.9
Haryana	NR	28.45001	77.01999	#####	130.3
Rajasthan	NR	26.45	74.63998	#####	234.1
Delhi	NR	28.66999	77.23	#####	85.8
UP	NR	27.59998	78.05001	#####	313.9
Uttarakha	NR	30.32041	78.05001	#####	40.7
HP	NR	31.10003	77.1666	#####	30
J&K	NR	33.45	76.24	#####	52.5
Chandigar	NR	30.72	76.78001	#####	5
Chhattisgar	WR	22.09042	82.15999	#####	78.7
Gujarat	WR	22.2587	71.1924	#####	319.5
MP	WR	21.30039	76.13002	#####	253
Maharash	WR	19.25023	73.16017	#####	428.6
Goa	WR	15.492	73.818	#####	12.8
DNH	WR	20.26658	73.01662	#####	18.6
Andhra Pr	SR	14.75043	78.57003	#####	164.6
Telangana	SR	18.1124	79.0193	#####	204.2
Karnataka	SR	12.57038	76.92	#####	206.3
Kerala	SR	8.00172	76.56000	#####	77.7

From the <https://naanmudhalvan.smartinternz.com/> website, after signing in by using login credentials, you should follow the steps down below

Projects

Access Resources

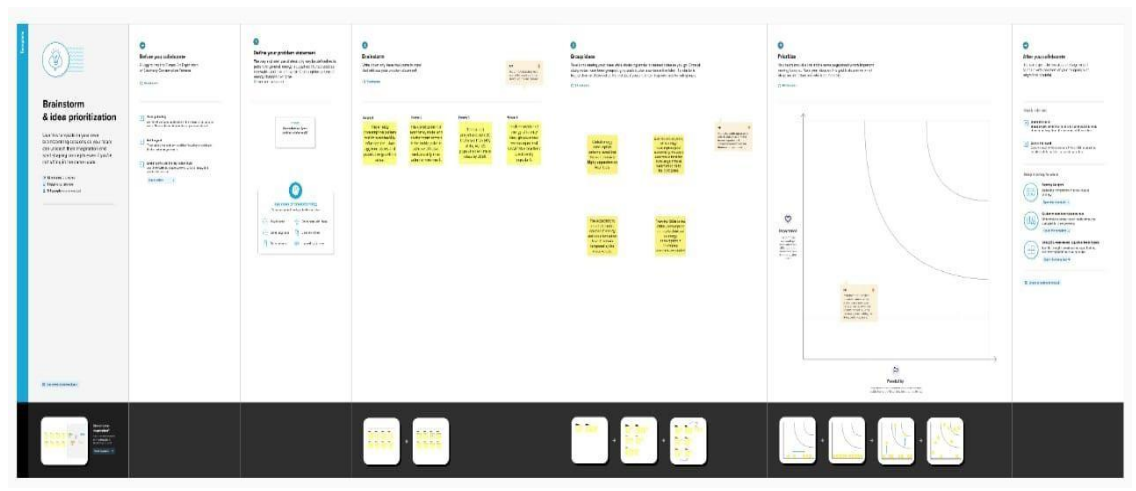
Guided Projects

Go To Workspace

Download Project Manual (to access the detailed steps to complete the project)

Then, clicking “Data Collection & Extraction” From Database, you can download the csv file of data for your project from “Collect the Dataset” down below.

○ BRAINSTORMING & IDEA PRIORITIZATION



By creating a new account in MURAL website <https://app.mural.co/signin> you can do the task of brainstorming and idea prioritization for the given project topic ‘Plugging into the Future: An Exploration

of Electricity Consumption Patterns’. You can follow the guided steps below.

- Log into the MURAL website
- Search for ‘BRAINSTROMING AND IDEA PRIORITIZATION’ in the templates
- Once it pops up, after previewing, create a new mural template and start working on your topic.
- After completing the template, download it as a pdf file for future reference.

○EMPATHY MAP

Empathy map can also be created for your topic same as the brainstorming by following the similar steps

- Search for ‘Empathy Map’ in the templates
- Create new mural template
- Do the empathy map task for your project topic

- Download it as pdf file once it's completed

Template

Empathy map

Electricity is a very important natural resources. It should be saved because it's not at all free. Energy conservation is the effort made by us to reduce the consumption of energy by using less of an energy service or using renewable sources of energy.

[Share template feedback](#)

Says

What have we heard them say?
What can we imagine them saying?

Electricity have become a basic human need in our modern society

Thinks

What are their wants, needs, hopes, and dreams? What other thoughts might influence their behavior?

More than a billion people around the world still lack access to electricity; more receive poor and inadequate supplies.

For the unserved and underserved, there can be significant implications for healthcare, food, education and business.

Financially self-sustaining electricity access initiatives cannot be supported without successful development that underpins demand for electricity services.

Governments can harmonize national policies for electricity access and create more synergies.

In most developing countries, electricity planners look to economic growth forecasts to project future demand for electricity.

Governments can build demand assessments into periodic national and subnational surveys.

Innovative consumer finance models are emerging to provide affordable electricity to consumers who want reliable services.

To expand electricity services to people without access, we need to look beyond the conventional technology-centric approach and adopt a more holistic perspective.

Does

What behavior have we observed?
What can we imagine them doing?

Electricity have become a basic human need in our modern society

Feels

What are their fears, frustrations, and anxieties? What other feelings might influence their behavior?

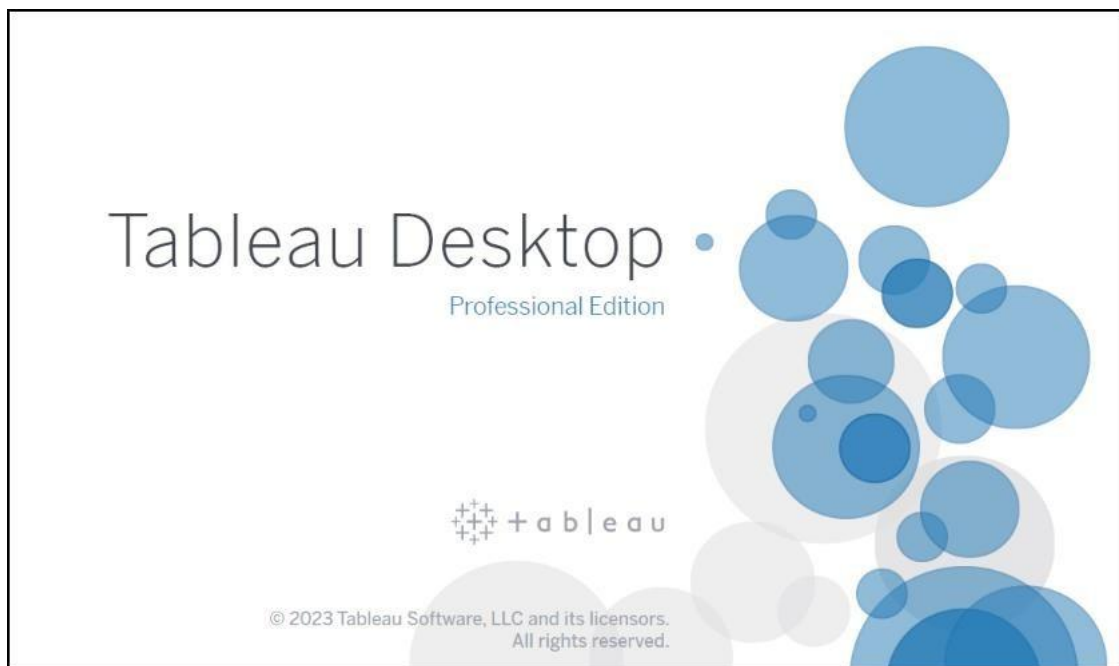
Need some inspiration?

See a finished version of this template to kickstart your work.

[Open example](#)

○TABLEAU

First of all, download the Tableau Desktop version on your PC. Then register by filling in the details required and start a free trail for 14 days or else you can use the product usage key if you had it.



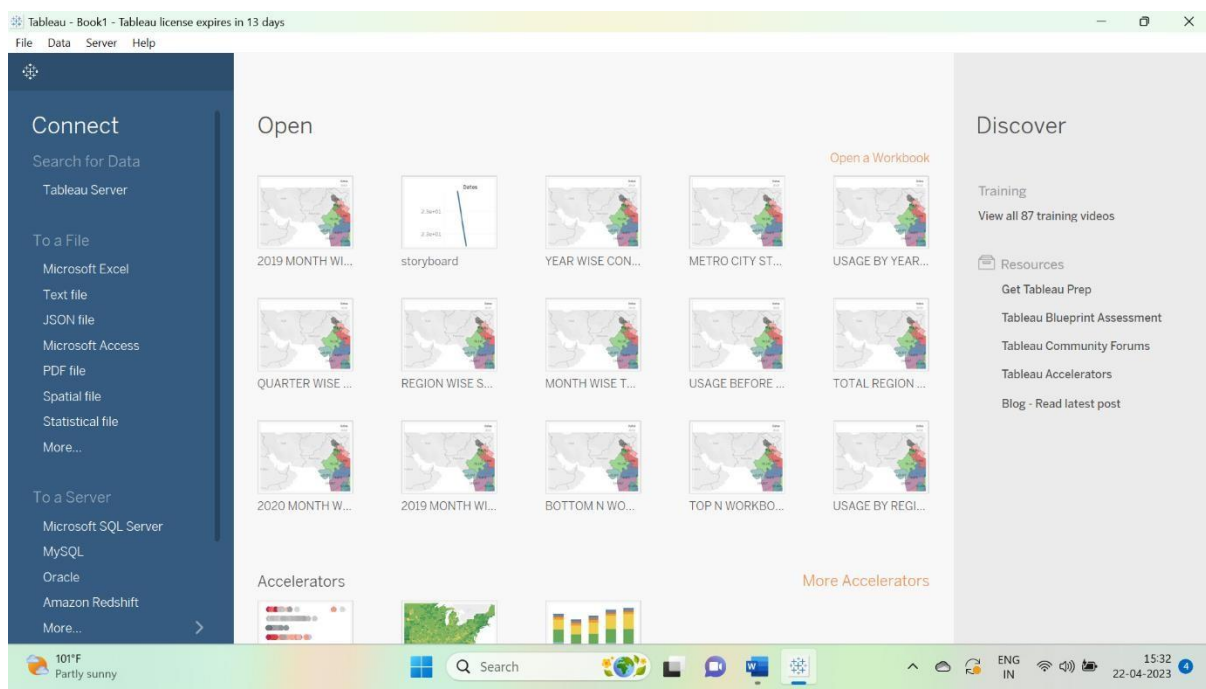
Then go to 'Microsoft Excel' to get started on the dashboard and story for your project.

Once the data sheet, you can start creating sheets as per instructed on your project manual videos under the data visualization section.

After completing the sheets, you can get started with your Dashboard by the instructions on the video in project manual

Once you are done with the dashboard, you can start creating your story on the Tableau as per the instructions in the video

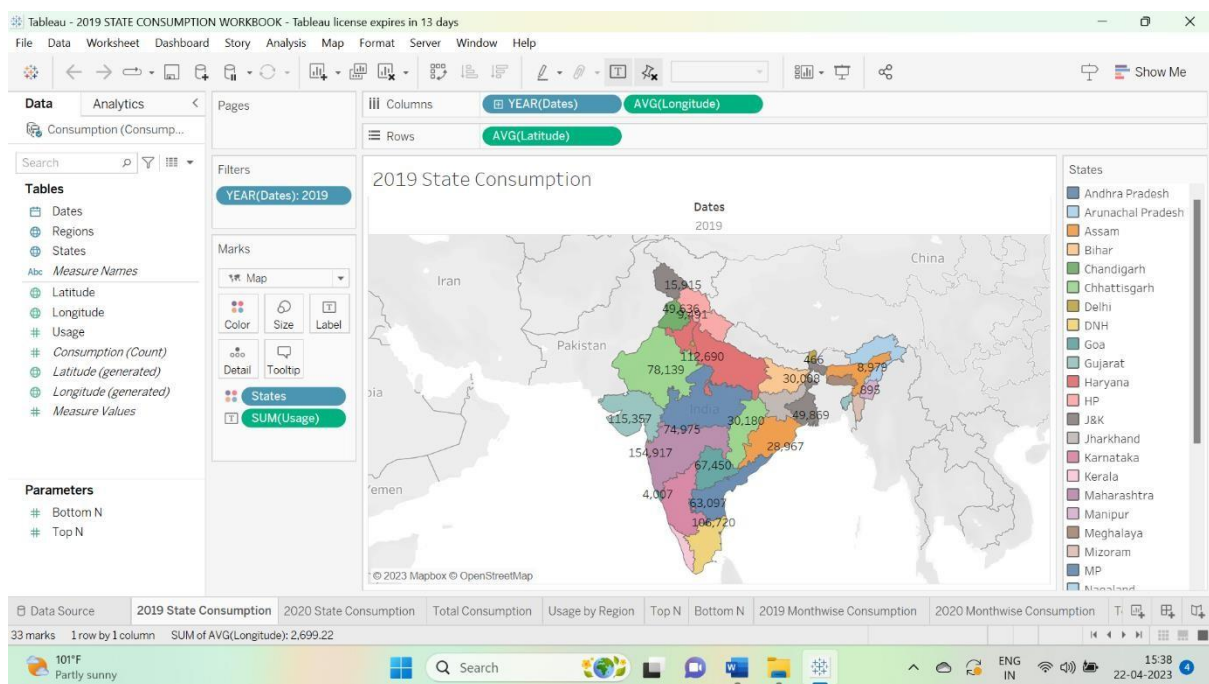
Once you are done with data visualization, dashboard and story, you have to extract the data and save the workbook in Tableau and in the end, this is how your Tableau app will look like.



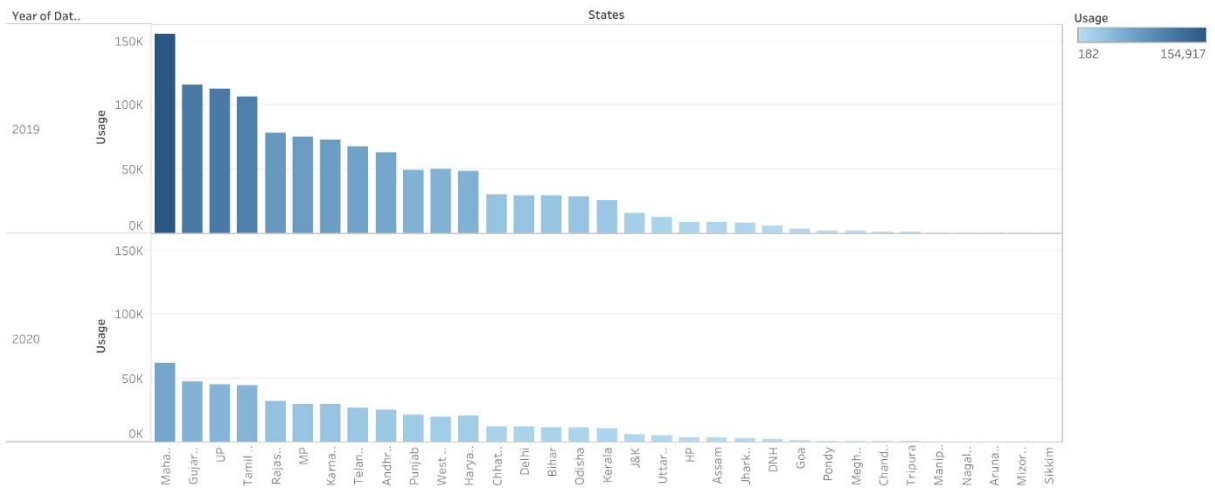
OWEB INTEGRATION

Now, you have to get started on publishing your data visualization, dashboards and story boards on <https://public.tableau.com/>

In this website, you have to create a new account and sign in using your email id and password created.



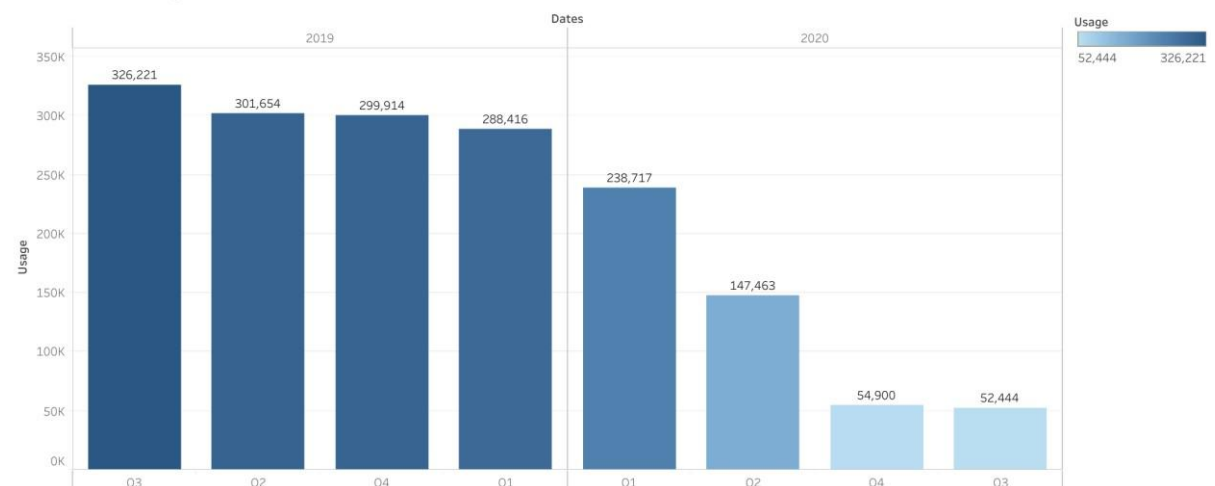
Total Consumption



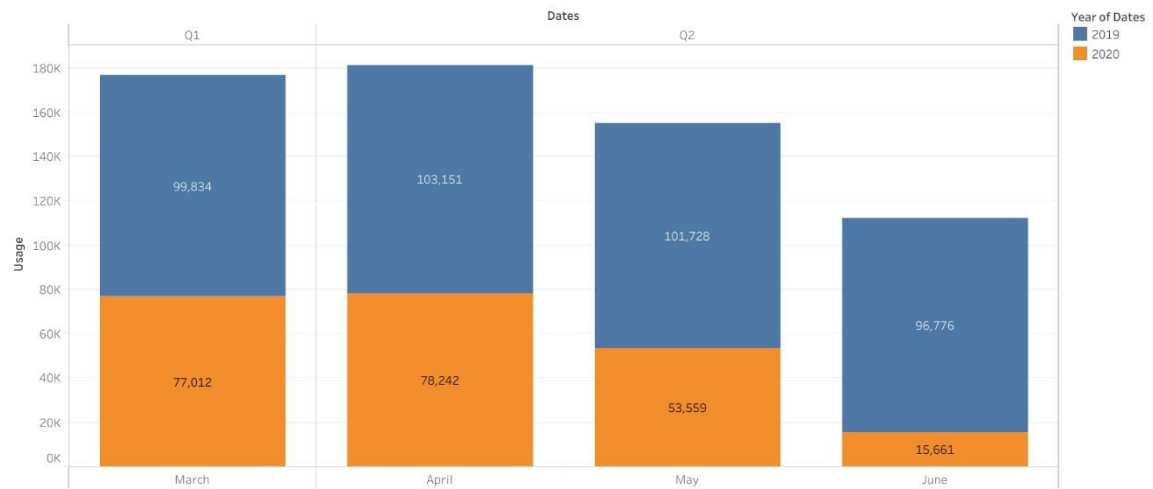
Top N



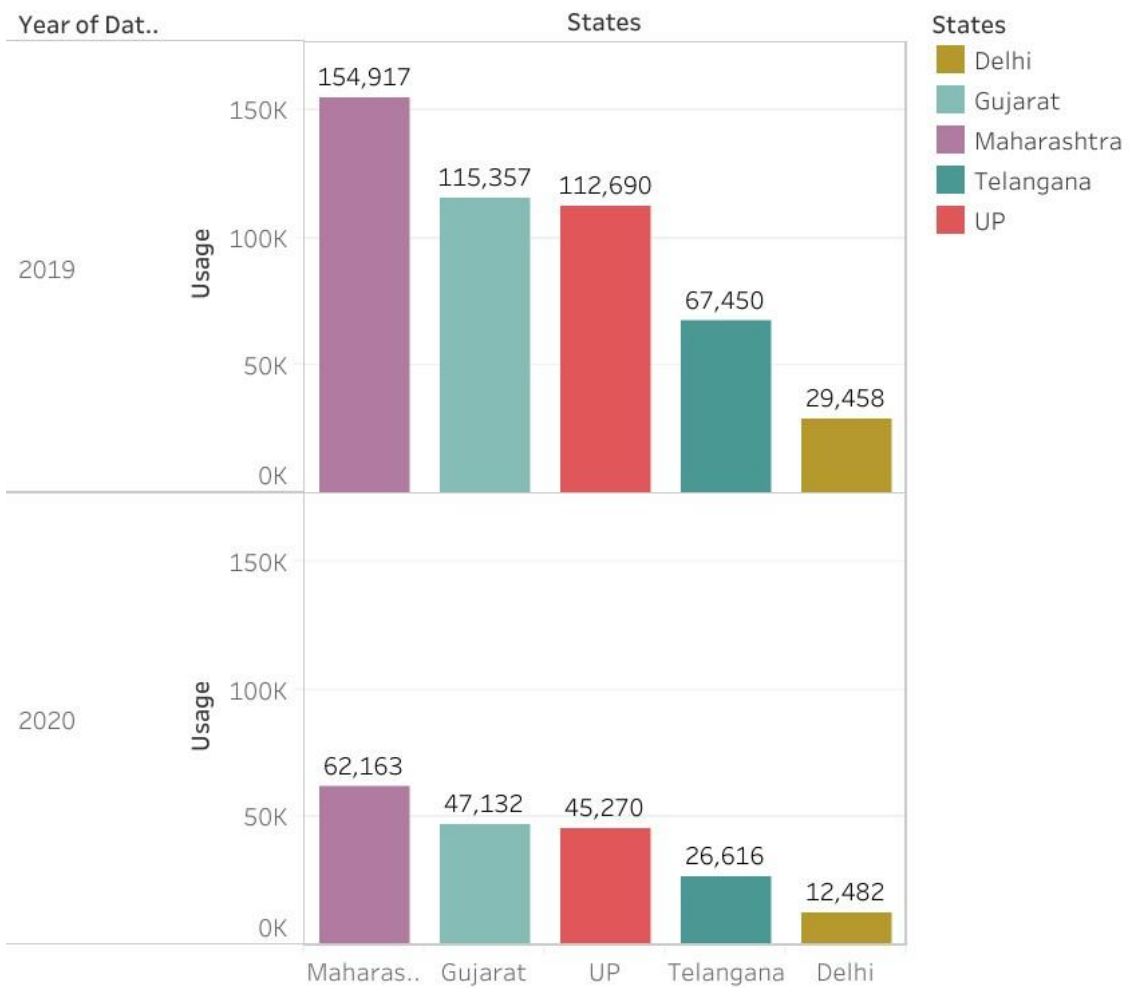
Quarter Wise Usage



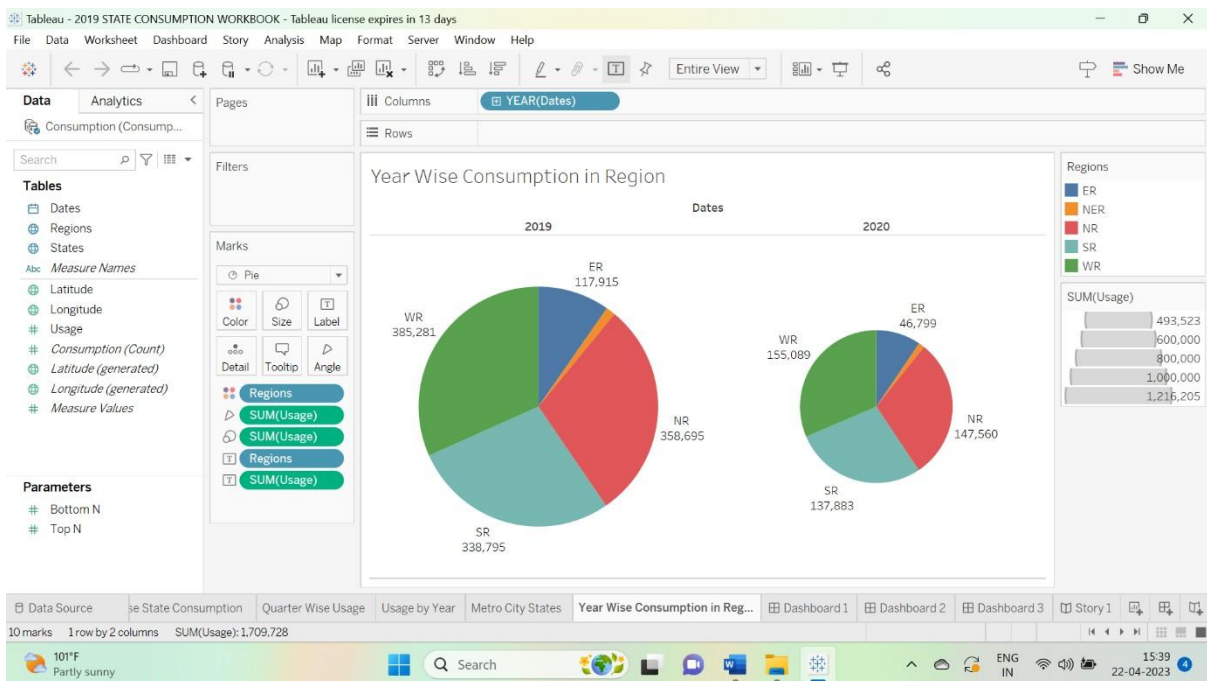
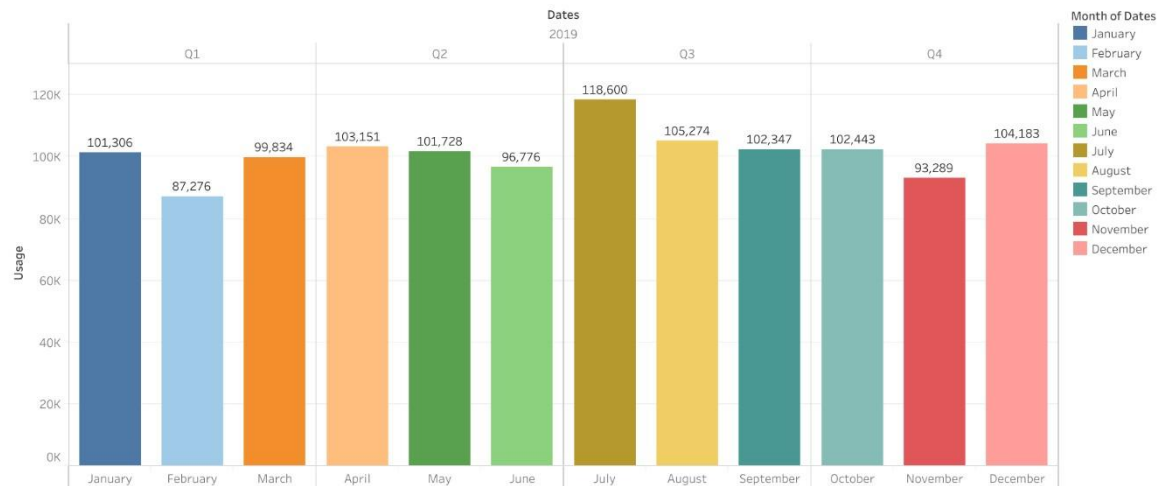
Usage before and during Lockdown



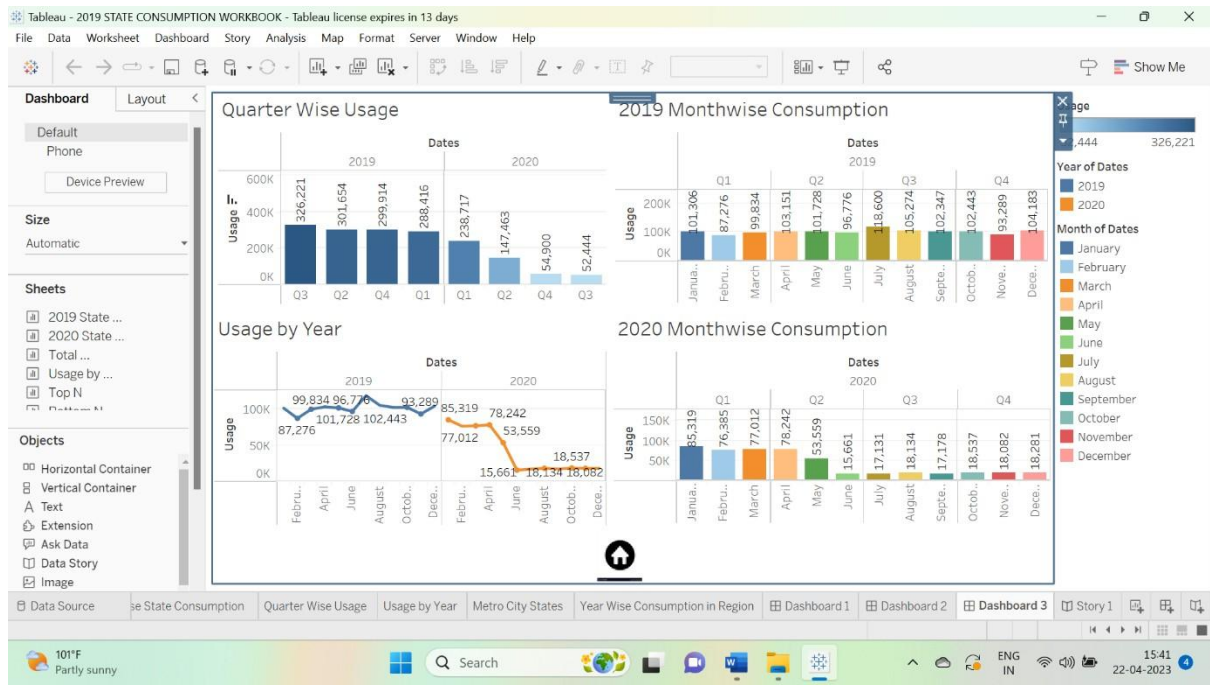
Metro City States



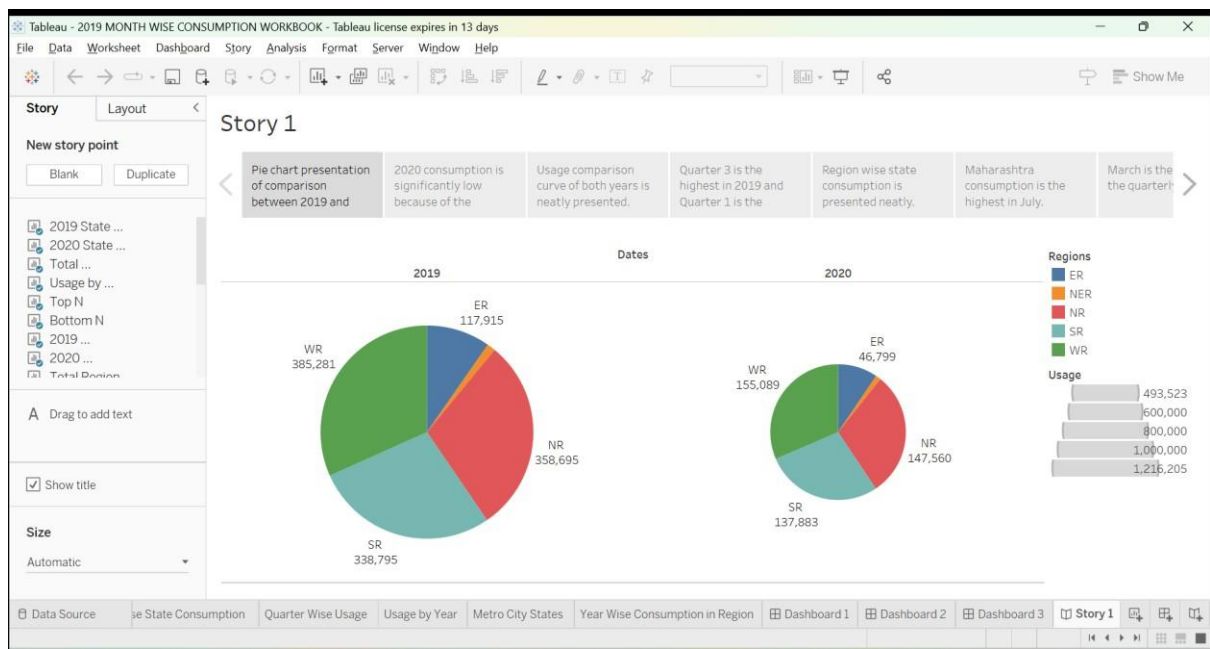
2019 Monthwise Consumption



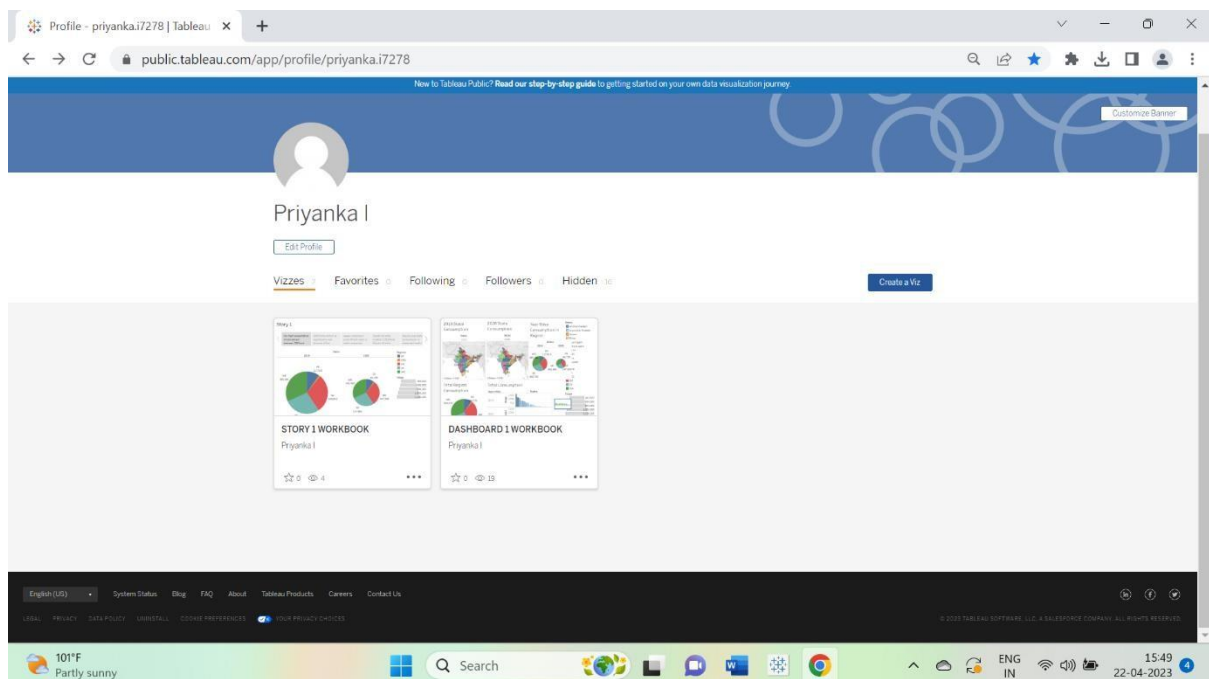
DASHBOARDS



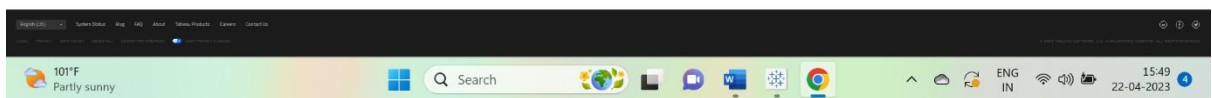
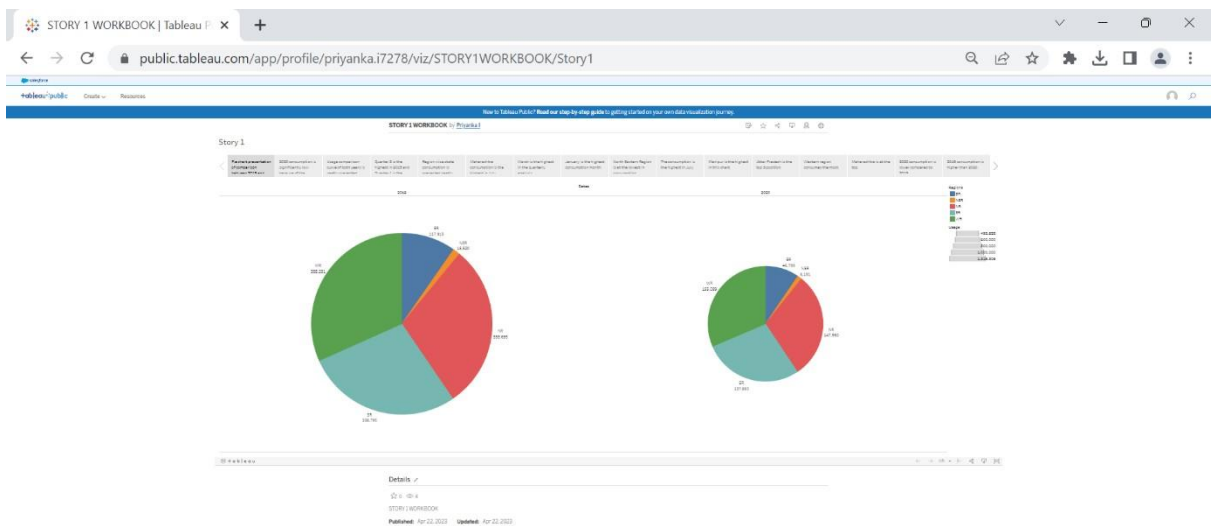
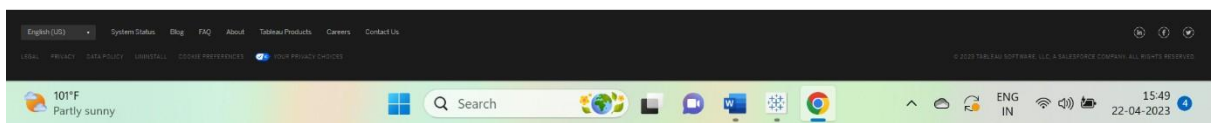
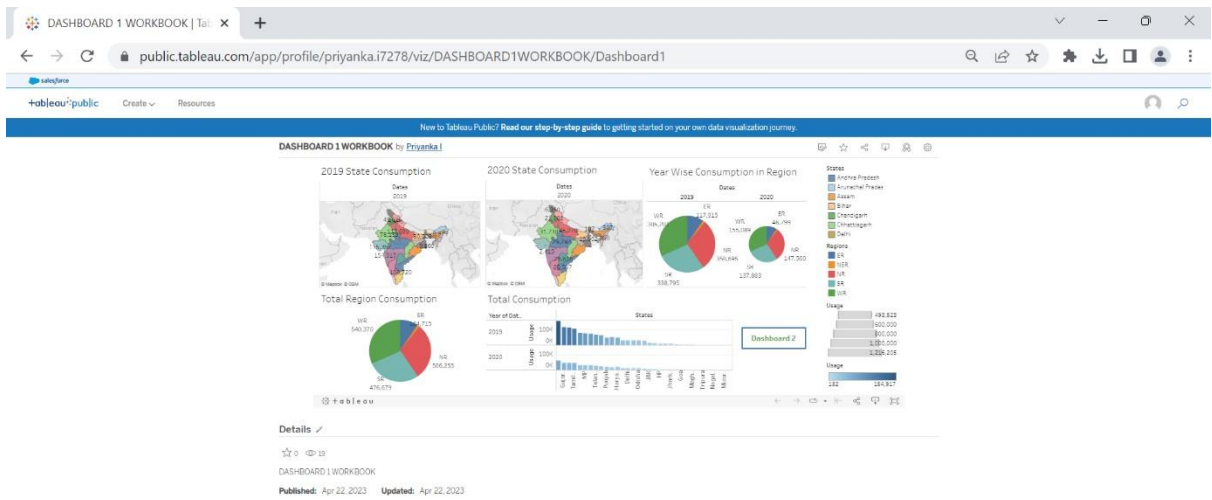
OSTORY



Now, open the dashboard and story workbook created on Tableau and start publishing the work on the given website using the share icon, then click 'connect' and save the workbook by giving it a title. After publishing dashboard and story on tableau public, this is how it will look like.



The pictures below are the published workbook of dashboard and story of the project shown on the above profile.



RESULTS & CONCLUSION

There was a considerable increase in the peak domestic consumption, as the peak load reached 3320 MW in 2017 with an annual increment rate of 4.9%. Regarding energy efficiency, the value of total electrical energy losses reached 13% in 2017; around 90% of this loss occurred in the electrical distribution stage.

Geographical distribution of the household electrical power shows that the east and middle parts of Amman have low consumption levels compared to the west residential parts. The energy consumption pattern has an inverse relation with the population distribution, family size, and building characteristics in the city. This is clearly identified by addressing the downtown region that has the lowest energy consumption and the highest-density population, while the western part has the highest energy consumption and low-population density. These variations can be referred to as differences in social and economic behaviours of inhabitants in both high-density and low-density population areas.

This analysis reflects the influence of several factors that should be taken into account in energy sustainability strategies. Energy consumption is

influenced by the characteristics of households which include building size, household income, total energy cost, and building characteristics (e.g., building design, age, location, and using thermal insulation system for buildings).

○ APPENDIX

SOURCE CODE

file:///C:/Users/OTHERS/AppData/Local/Temp/Temp1_Electricity%20Consumption%20Analysis-20230422T122821Z-001.zip/Electricity%20Consumption%20Analysis/index.html

TABLEAU PROFILE LINKS FOR

DASHBOARD

https://public.tableau.com/views/DASHBOARD1WORKBOOK/Dashboard1?:language=en-US&:display_count=n&:origin=viz_share_link

STORY

https://public.tableau.com/views/STORY1WORKBOOK/Story1?:language=en-US&:display_count=n&:origin=viz_share_link